

MD ARMAN

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Education

Indian Institute of Technology Delhi May 2026 *Bachelor of Technology in Material Science and Engineering New Delhi, India*

Experience

Skolist Dec 2025 – May 2026 *Founding Engineer Remote*

- Served as sole engineer on an AI-powered EdTech startup, owning architecture, sprint planning, and deployment end-to-end from 0 → 1.
- Built and shipped a full-stack platform (Next.js, Node.js, PostgreSQL) with REST APIs, normalized database schemas, and an adaptive assessment engine.
- Designed a personalized analytics and recommendation engine that flagged individual student learning gaps and surfaced targeted content to close them.
- Applied System Design principles (caching, query optimization, modular service boundaries) to keep the platform responsive as the assessment engine scaled.

Projects

HandToHand — P2P Campus Logistics & Economy Platform

- Architected a decentralized assistance network for IIT Delhi using **Next.js, Node.js, PostgreSQL**, enabling P2P errands, item lending, and smart campus logistics — demonstrating full-stack ownership.
- Engineered a geofence-based matching engine using **Redis GEOSEARCH** for event-driven notifications, optimizing battery usage by eliminating continuous GPS polling.
- Built a Real-time Coin Economy with **Prisma** transactions and escrow system for secure P2P rewards; implemented Intentional Trip Broadcasting via **Socket.IO** and **BullMQ** with real-time slot management.
- Integrated Google OAuth (IITD domain-restricted) and a reputation-scoring algorithm based on completion rates and user ratings to ensure a high-trust campus environment.

TubeTrend — Autonomous AI YouTube Content Strategist

- Built a full-stack AI agent using **Next.js, Node.js, and Pinecone** that automates content research via a multimodal RAG pipeline; fetched transcripts via YouTube Data API with semantic vector embeddings.
- Integrated **Groq LLM (Llama 3.3)** to generate high-retention scripts, SEO-optimized descriptions, and content summaries based on viral trend analysis.
- Implemented Context-Aware Memory for multi-video analysis enabling cross-playlist/channel querying with source-cited AI responses; designed a responsive content performance forecasting dashboard.

Technical Skills

Languages & Frameworks: TypeScript/JavaScript, Python, HTML, CSS, Next.js, React, Redux, Node.js, Express, Django, Tailwind CSS

Databases & Backend: PostgreSQL, MySQL, MongoDB, Redis, Pinecone (Vector DB), Prisma, Socket.IO, BullMQ, Firebase, Docker, AWS EC2

AI/ML Tools: LangChain, RAG, Groq LLM, HuggingFace Embeddings, Google ADK, Prompt Engineering, MCP

CS Fundamentals: Data Structures & Algorithms, System Design, OOP, JWT Authentication, OAuth 2.0, Git/GitHub